

SEE 2083: Optional Mathematics

Complete Formula Sheet

1. Algebra

Functions:

Composite Function: $f \circ g(x) = f(g(x))$

Inverse Function: If $y = f(x)$, then $x = f^{-1}(y)$.

Polynomials:

Remainder Theorem: If a polynomial $f(x)$ is divided by $(x - a)$, the remainder R is $f(a)$.

Factor Theorem: If $f(a) = 0$, then $(x - a)$ is a factor of $f(x)$.

2. Matrix & Determinants

For a matrix $A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$:

- **Determinant:** $|A| = ad - bc$
- **Inverse Matrix (A^{-1}):** $A^{-1} = \frac{1}{|A|} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$, (Exists if $|A| \neq 0$)

Cramer's Rule (For $a_1x + b_1y = c_1$ and $a_2x + b_2y = c_2$):

$$D = \begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}, D_x = \begin{vmatrix} c_1 & b_1 \\ c_2 & b_2 \end{vmatrix}, D_y = \begin{vmatrix} a_1 & c_1 \\ a_2 & c_2 \end{vmatrix}$$

$$x = \frac{D_x}{D}, \quad y = \frac{D_y}{D}$$

3. Coordinate Geometry

Angle Between Two Lines ($y = m_1x + c_1$ and $y = m_2x + c_2$):

$$\tan \theta = \pm \frac{m_1 - m_2}{1 + m_1 m_2}$$

Parallel Condition: $m_1 = m_2$

Perpendicular Condition: $m_1 \times m_2 = -1$

Circle Equations:

- Standard Form (center (h, k) , radius r): $(x - h)^2 + (y - k)^2 = r^2$
- General Form: $x^2 + y^2 + 2gx + 2fy + c = 0$ (Center: $(-g, -f)$, $r = \sqrt{g^2 + f^2 - c}$)
- Diameter Form: $(x - x_1)(x - x_2) + (y - y_1)(y - y_2) = 0$

4. Trigonometry

Multiple Angles:

- $\sin 2A = 2 \sin A \cos A = \frac{2 \tan A}{1 + \tan^2 A}$
- $\cos 2A = \cos^2 A - \sin^2 A = 2 \cos^2 A - 1 = 1 - 2 \sin^2 A = \frac{1 - \tan^2 A}{1 + \tan^2 A}$
- $\tan 2A = \frac{2 \tan A}{1 - \tan^2 A}$
- $\sin 3A = 3 \sin A - 4 \sin^3 A$
- $\cos 3A = 4 \cos^3 A - 3 \cos A$

Transformation Formulas:

- $2 \sin A \cos B = \sin(A + B) + \sin(A - B)$
- $2 \cos A \sin B = \sin(A + B) - \sin(A - B)$
- $2 \cos A \cos B = \cos(A + B) + \cos(A - B)$
- $2 \sin A \sin B = \cos(A - B) - \cos(A + B)$
- $\sin C + \sin D = 2 \sin \left(\frac{C+D}{2} \right) \cos \left(\frac{C-D}{2} \right)$
- $\cos C + \cos D = 2 \cos \left(\frac{C+D}{2} \right) \cos \left(\frac{C-D}{2} \right)$

5. Vectors

Dot Product: $\vec{a} \cdot \vec{b} = |\vec{a}| |\vec{b}| \cos \theta = a_1 b_1 + a_2 b_2$

Vector Geometry:

- Mid-point Formula: $\vec{m} = \frac{\vec{a} + \vec{b}}{2}$
- Section Formula (Internal): $\vec{p} = \frac{m\vec{b} + n\vec{a}}{m+n}$
- Section Formula (External): $\vec{p} = \frac{m\vec{b} - n\vec{a}}{m-n}$

6. Statistics

Measures of Dispersion:

- Mean ($\bar{x}$): $\frac{\sum fm}{N}$
- Mean Deviation: $MD_{\bar{x}} = \frac{\sum f|m-\bar{x}|}{N}$
- Standard Deviation (σ): $\sqrt{\frac{\sum f(m-\bar{x})^2}{N}}$ OR $\sqrt{\frac{\sum fm^2}{N} - \left(\frac{\sum fm}{N} \right)^2}$
- Coefficient of Variation: $CV = \frac{\sigma}{\bar{x}} \times 100\%$